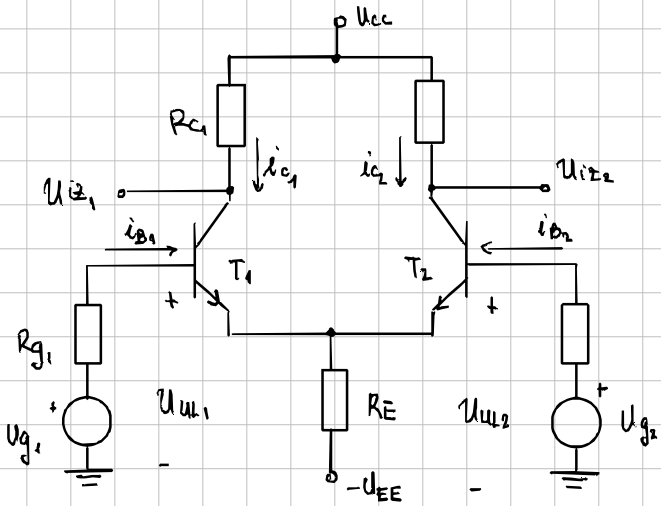
Primjer 1.2.) Za dif. P. iz pr. 1.1. za asimet ulaz izračunati naponska pojačanja zajedničkog i diferencijskog signala A_{vz} i A_{vd}, te faktor potiskivanja ρ.

$h_{fe} = 100$, $U_T = 25\,mV$



Statika: $U_\gamma = 0.7\,V$

$U_{CC}, U_{EE} = 15\,V$

$I_{BQ} = 15{,}7\,\mu A$

$I_{CQ} = 1{,}57\,mA$

$U_{CEQ} = 13{,}5\,V$

k.s. izmjenično

$R_{C1} = R_{C2} = 1{,}5\,k\Omega$

$R_{g1} = R_{g2} = 500\,\Omega$

$R_E = 4{,}5\,k\Omega$

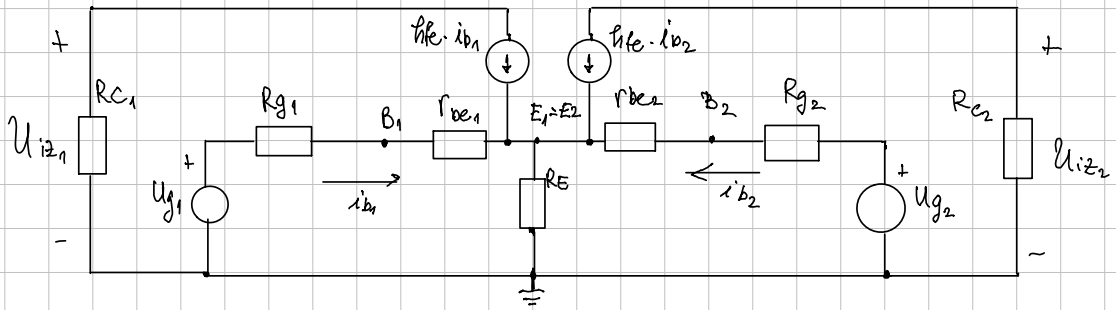
Dinamika

$$U_{g1} = U_z - \frac{U_d}{2}$$

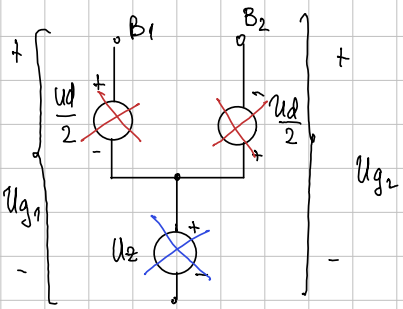
$$U_{g2} = U_z + \frac{U_d}{2}$$

Dinamički par: $r_{be} = \frac{U_T}{I_{BQ}} = 1.59\,k\Omega$

— uzemlji istosmjerno



→ metodom superpozicije gledamo 2 slučaja → U_z i U_d

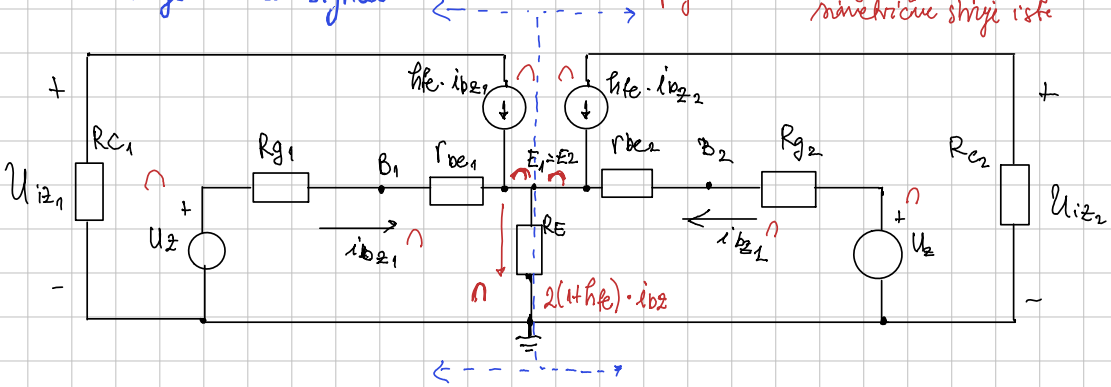


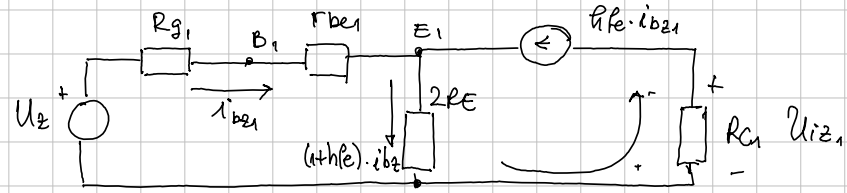
1. slučaj → zajednički signal = signal smetnji (taj je istofazni)

2. slučaj → diferencijski signal

Zajednički signal:

simetričan spoj → možemo gledati pola jer su simetrične struje iste

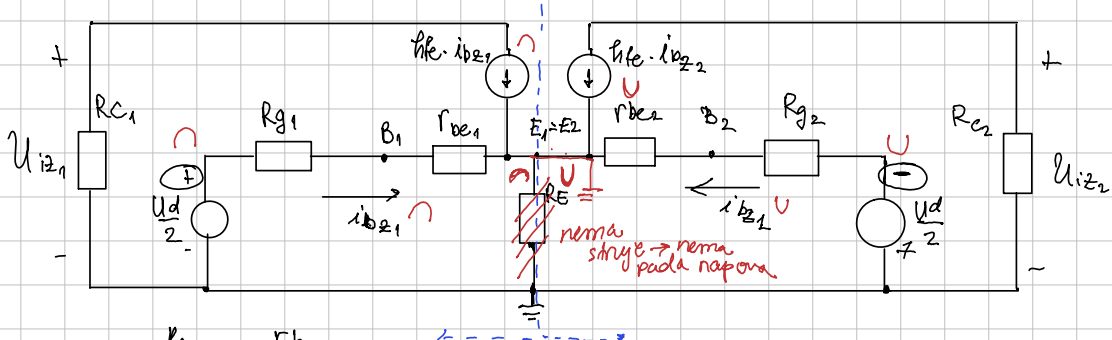


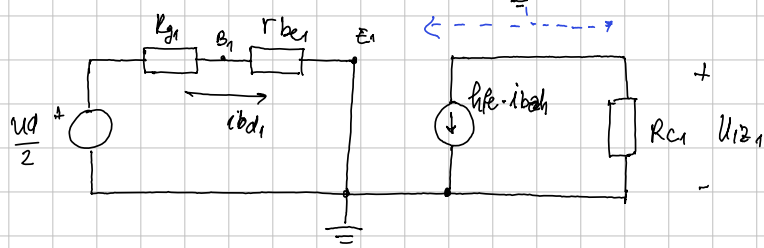


$$A_{vz} = \frac{u_{iz}}{u_z} = \frac{-h_{fe}\cdot \cancel{i_{b1}}\cdot R_{c1}}{(1+h_{fe})\cancel{i_{b1}}\cdot(R_{g1}+r_{be1}+2R_E)} = \frac{-h_{fe}\cdot R_{c1}}{(1+h_{fe})(R_g+r_{be1}+2R_E)} = \dots = -0.134$$

gušenje ✓

Diferencijski signal:





pojačanje bez degeneracije ✓

$$A_{vd} = \frac{u_{iz}}{u_d} = \frac{u_{iz}}{2\cdot\frac{u_d}{2}} = \frac{-h_{fe}\cdot \cancel{i_{bd1}}\, R_{c1}}{2\cdot \cancel{i_{bd1}}(R_{g1}+r_{be1})} = \frac{-1}{2}\cdot\frac{h_{fe}\cdot R_{c1}}{R_{g1}+r_{be1}} = \dots = -35.9$$

Faktor potiskivanja: $\rho = ?$

$$\rho = \left|\frac{A_{vd}}{A_{vz}}\right| = \left|\frac{-35.9}{-0.134}\right| = 268 \rightarrow \rho_{dB} = 20\log(\rho) = 48.5\,dB$$

Primjer 3.) $U_{g_1} = U_{g_{1m}} \sin\omega t$ i $U_{g_2} = U_{g_{2m}} \sin\omega t$

izračunaj izlazni napon $U_{iz} = U_{iz_2}$?

a) $U_{g_{1m}} = -5mV$ $\quad U_{g_{2m}} = 5mV$

$U_z = \frac{U_{g_1} + U_{g_2}}{2}$ $\quad U_d = U_{g_2} - U_{g_1} \rightarrow U_{d_m} = 10mV$

$U_z = \emptyset$ $\quad U_d = 10 \sin\omega t \, [mV]$

$A_{vz} = -0.134$ $\quad A_{vd} = -35.9$

$U_{iz_{2m}} = A_{vz_2} \cdot U_{zm} + A_{vd_2} \cdot U_{dm}$

$= -0.134 \cdot \emptyset + (-35.9) \cdot 10mV \Rightarrow U_{iz_{2m}} = -359mV$

b) $U_{g_{1m}} = 20mV$ $\quad U_{g_{2m}} = 30mV$

$U_{z_m} = \frac{U_{g_1} + U_{g_2}}{2} = 25mV$ $\quad U_{dm} = U_{g_{m2}} - U_{g_{m1}} = 10mV$

$U_{iz_{2m}} = A_{vz_2} \cdot U_{zm} + A_{vd_2} \cdot U_{dm} = -0.134 \cdot 25mV + (-35.9) \cdot 10mV$

$= -4.1 - 359$

$U_{iz_{2m}} = -362.35mV$

prigušio smetnje da bude čišć na izlazu